

Nonhomogeneous Equations & Undetermined Coefficients

ANSWER KEY



Complementary function and particular solution

- 1 For $y'' - 3y' + 2y = 4e^{3x}$, find the complementary function y_c .
 $r^2 - 3r + 2 = (r - 1)(r - 2) = 0 \Rightarrow r = 1, 2$. So $y_c = C_1e^x + C_2e^{2x}$.
- 2 Explain why the general solution of a nonhomogeneous linear equation is $y = y_c + y_p$.
If y_p is any particular solution and y_c solves the associated homogeneous equation, then $y_c + y_p$ solves the nonhomogeneous equation (by linearity/superposition) and carries enough arbitrary constants to match any initial condition — so it is the general solution.

Choosing a trial form (no resonance)

- 3 Find a particular solution of $y'' - 3y' + 2y = 4x$.
Try $y_p = Ax + B$: $y_p' = A$, $y_p'' = 0$. Substituting: $-3A + 2(Ax + B) = 4x \Rightarrow 2Ax + (2B - 3A) = 4x$. So $2A = 4 \Rightarrow A = 2$, and $2B - 3A = 0 \Rightarrow B = 3$. Thus $y_p = 2x + 3$.
- 4 Find a particular solution of $y'' + y' - 2y = 6e^{3x}$ (note $r = 3$ is not a root of $r^2 + r - 2 = 0$).
Try $y_p = Ae^{3x}$: $y_p' = 3Ae^{3x}$, $y_p'' = 9Ae^{3x}$. Substituting: $9A + 3A - 2A = 6 \Rightarrow 10A = 6 \Rightarrow A = \frac{3}{5}$. Thus $y_p = \frac{3}{5}e^{3x}$.

The modification rule (resonance)

- 5 Find a particular solution of $y'' - 3y' + 2y = e^{2x}$. (Note $r = 2$ IS a root of $r^2 - 3r + 2$.)
Since e^{2x} solves the homogeneous equation, try $y_p = Axe^{2x}$. Then $y_p' = Ae^{2x} + 2Axe^{2x}$ and $y_p'' = 4Ae^{2x} + 4Axe^{2x}$. Substituting: $(4Ae^{2x} + 4Axe^{2x}) - 3(Ae^{2x} + 2Axe^{2x}) + 2Axe^{2x} = Ae^{2x}$ (the xe^{2x} terms cancel). Setting $Ae^{2x} = e^{2x}$ gives $A = 1$. Thus $y_p = xe^{2x}$.

Combining forcing terms

- 6 Find a particular solution of $y'' - y = e^x + x$. (Roots of $r^2 - 1 = 0$ are $r = \pm 1$, so e^x resonates but x does not.)
For the e^x part (resonant): try Axe^x , giving $y_p'' - y_p = 2Ae^x = e^x \Rightarrow A = \frac{1}{2}$, so $y_{p1} = \frac{1}{2}xe^x$. For the x part: try $Bx + C$, giving $-(Bx + C) = x \Rightarrow B = -1, C = 0$, so $y_{p2} = -x$. Total: $y_p = \frac{1}{2}xe^x - x$.

Application: a driven oscillator

- 7 A forced oscillator satisfies $y'' + 4y = \sin 3t$ (forcing frequency $3 \neq$ natural frequency 2 — no resonance). Find a particular solution.

Try $y_p = A \sin 3t + B \cos 3t$: $y_p'' = -9A \sin 3t - 9B \cos 3t$. Substituting:
 $-5A \sin 3t - 5B \cos 3t = \sin 3t \Rightarrow A = -\frac{1}{5}, B = 0$. Thus $y_p = -\frac{1}{5} \sin 3t$.

- 8 Now solve $y'' + 4y = \sin 2t$, where the forcing frequency EQUALS the natural frequency (resonance). Find a particular solution.

Since $\sin 2t, \cos 2t$ solve the homogeneous equation, try $y_p = t(A \sin 2t + B \cos 2t)$. Computing y_p'' and substituting, the terms proportional to $t \sin 2t$ and $t \cos 2t$ cancel, leaving
 $y_p'' + 4y_p = 4A \cos 2t - 4B \sin 2t$. Setting this equal to $\sin 2t$ gives $A = 0, B = -\frac{1}{4}$. Thus $y_p = -\frac{1}{4} t \cos 2t$ — the amplitude grows linearly in t , the signature of resonance.